

A Fuzzy Multi-Criteria Group Decision-Making Model Based on Weighted Borda Scoring Method for Watershed Ecological Risk Management: a Case Study of Three Gorges Reservoir Area of China

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Abstract In watershed ecological risk management, a series of alternatives will be analyzed in terms of multiple complex criteria, and different stakeholders with conflicting risk attitudes will be involved, which means that ecological risk management decision-making is a process surrounded by a wide range of uncertainties derived from scarcity of data, lack of knowledge, deficiency of assumptions and lexical vagueness. Based on modified Borda scoring method, this paper discusses how to apply a fuzzy multi-criteria group decision-making (FMCGDM) model into such a process. Then, a two-stage, 12-step MCDM methodology is proposed to obtain the optimal alternative decided by multiple decision-makers (DMs) from a given alternatives set. Firstly, all DMs make their own independent choice respectively, then, all the independent conclusions are integrated by using their subjective/objective weights. A modified Borda method is followed to rate and rank the weighted alternatives; in which the one with the highest score will be selected as the final preferable option. This model is demonstrated to be applicable and reliable by an application in the ecological risk decision making process of Three Gorges Reservoir area located in the upper reaches of Yangtze River in China. Of course, this method can also be applied in other research fields of environmental management.

Keywords Ecological risk management · Multi-criteria decision-making · Fuzzy theory · Watershed management · Three Gorges Reservoir area

List of notation

$\mu(x)$ Degree of membership
 A_i Alternatives set; ($i = 1, 2, \dots, m$)

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$\tilde{X}$	Fuzzy set
$\tilde{x}$	Fuzzy number in fuzzy set $\tilde{X}$, in this paper, all the fuzzy numbers are expressed as triangular fuzzy numbers (TFNs)
$\tilde{x}_L, \tilde{x}_R$	Left fuzzy set and right fuzzy set of fuzzy number $\tilde{x}$
$\otimes$	operator
D_k	The k th decision maker; ($k = 1, 2, \dots, n$)
$\tilde{W}_k$	Relative importance (weight) of D_k in the decision makers team
$\tilde{P}^k$	Judgment matrix proposed by D_k
C_j	Criteria set of alternative ($j = 1, 2, \dots, L$)
$\tilde{P}_{ij}^k$	Element in judgment matrix ($\tilde{P}^k$), which represents the fuzzy judgment proposed by decision maker (D_k) to criterion (C_j) of alternative (A_i).
$\tilde{\lambda}_{ij}^k$	Normalized $\tilde{P}_{ij}^k$
$\tilde{w}_k$	Fuzzy weighting vector assigned by decision maker (D_k)
$\tilde{w}_{jk}$	Element of $\tilde{w}_k$, denotes fuzzy weight for criterion (C_j) assigned by decision maker (D_k)
$\tilde{r}_{ij}^k$	Weighted $\tilde{\lambda}_{ij}^k$ in judgment matrix ($\tilde{P}^k$)
$\tilde{B}$	A fuzzy set, representing the ideal optimal situation derived from alternatives set A
$\tilde{I}_k^+$	Fuzzy positive ideal alternative vector designated by decision maker (D_k)
$\tilde{I}_{jk}^+$	Element of alternative vector ($\tilde{I}_k^+$), representing the optimal value of criterion (C_j)
$\tilde{I}_k^-$	Fuzzy negative ideal alternative vector designated by decision maker (D_k)
$\tilde{I}_{jk}^-$	Element of alternative vector ($\tilde{I}_k^-$), representing the worst value of criterion (C_j)
$\tilde{R}_{jk}^+$	Weighted $\tilde{I}_{jk}^+$
$\tilde{R}_{jk}^-$	Weighted $\tilde{I}_{jk}^-$
F_{ik}^+	Euclidean distance between each alternative (A_i) and weighted $\tilde{I}_k^+$
F_{ik}^-	Euclidean distance between each alternative (A_i) and weighted $\tilde{I}_k^-$
$d(\tilde{x}, \tilde{y})$	Distance between two fuzzy numbers $\tilde{x}$ and $\tilde{y}$
β_k	Risk tolerance index of decision maker (D_k)
C_{ik}	relative closeness between decision alternative (A_i) and the positive ideal alternative ($\tilde{I}_k^+$)

1 Introduction

1.1 Background

Ecological risk assessment is a process that evaluates the likelihood that adverse ecological effects may occur or are occurring as a result of exposure to one or more stressors (U.S. EPA 1992). The process is used to systematically evaluate and organize data, information, assumptions, and uncertainties in order to help understand and predict the relationships between stressors and ecological effects in a way that is useful for environmental decision-making (U.S. EPA 1998), which means that a rational and reasonable risk management and decision-making process

is not only an integral part, but also the ultimate purpose of a complete ecological risk assessment process. One of the advices to USEPA on advancing the science and application of ecological risk assessment in environmental decision-making provided by the Science Advisory Board's (SAB) Ecological Processes and Effects Committee (EPEC) emphasizes that scientific decision-making should be looked upon as the ultimate goal, risk management alternatives should be formulated and evaluated at multiple scales and each of the alternative should be communicated to different interested stakeholder groups (SAB 2007).

Risk decision-making is a process of developing and selecting management programs and coping strategies in an ecological environment faced with risks. It is a benefit and cost trade-off between each kind of proposals. An ideal risk decision-making is a process of selecting the best one in a set of prioritized alternatives. But in reality, the optimization process is often very difficult to carry out, because the consequences of the risk and its possibility of occurrence and management cost usually are not uniform. For example, chemical plant explosion belongs to the category of small probability events in ecological risk assessment; the probability of river pollution derived from fire wastewater of chemical plant explosion is even smaller under normal circumstances. However, it had been proved that these small probability events can cause huge environmental pollution, e.g., the consequences of water pollution accident of Songhua River in Northeast China in 2005, which released 100 tons of benzene and nitrobenzene into the river, were disastrous. The water supply of in Harbin City was cut off and the pollutants even flowed into Russia, causing great financial losses and ecological damage. Initially, when the fire wastewater pollution just happened, at least two alternatives are proposed to the policy makers of Jilin municipality, Jilin Province; one is to take in situ remediation and control the scope of the pollution within Jilin Province, which will make local authority of Jilin municipality to pay the high expense of treatment; the other is to open the sluice gate of a upper reservoir to dilute the pollutants and wash the wastewater down to the lower reach. Unfortunately, without the assistance of corresponding specialists or a decision support system, the decision makers' preference to cheaper expense of the second option prevailed, which was proved absolutely wrong by the extension of pollution. And that is why policy makers need to carefully balance between the sustainable consequences of a potential accident and its possibility and expenses to make a feasible decision.

By now, no matter the fundamental theory and methodology research or the practical management application, the related competent authorities of China are still lack of sufficient understanding and corresponding preparation for watershed ecological risk management.

Most especially, the ecological risk managers of China are facing a transition process from experiential decision-making to modern scientific decision-making. A comprehensive, systematical and scientific decision support system (DSS) is urgently needed for Chinese decision makers to support the ecological risk management. Meanwhile, along with the ever-increasing complexity of environmental management and risk decision-making, unceasing expansion of the scope of management, the diversification of management level and participants, managers might take a comprehensive consideration based on mathematical models between a number of conflicting and inter-constraint factors or programs described by different evaluation criteria.

Based on the above reasons, this article introduces a modified FMCGDM method. Taking the risk assessment and management practice of Three Gorges reservoir area located in China's Yangtze River as a case, this paper validates the effectiveness of the new methodology.

1.2 The Applications of MCDA in Environment Management and Risk Decision-Making

Two basic types of risk management to consider are quantitative risk analysis and qualitative risk analysis. Both of them are based on a systemic approach to risk analysis. Qualitative risk analysis can be used to improve awareness of information systems security problems and the posture of the system being analyzed (James 1999). It is based on the use of qualitative information and cognitive map to describe the interaction between the different risk factors. A qualitative risk analysis is scenario-based (Miller 2002). It is performed when the organization requires a risk assessment and management be performed in a relatively short time or to meet a small budget, a significant quantity of relevant data is not available, or the persons performing the assessment don't have the sophisticated mathematical, financial, and risk assessment expertise required (Anderson 2005). Contrarily, in a quantitative risk analysis, the main elements have to be quantified. That is, the disutilities of the risk scenario and the probability have to be provided in a quantitative way. It is based substantially on independently objective processes and metrics and requires an accordingly increased degree of effort be placed in deterring the cost values and an increasing amount of effort be placed into the risk calculations (James 1999). Multi-criteria decision-making analysis (MCDA), as a hybrid methodology of a selected combination of qualitative and quantitative method, has been broadly used in risk decision-making process.

The MCDA methods are most commonly used in the field of water resources management in Chinese environmental management, which mainly focus on the research of decision support system. Those studies were mainly targeted at the influence of individual water resource development project on economic, social and environment, or at the environmental planning and management problems of a single catchment. So far, there aren't many integrated cases of ecological risk management with MCDA models in China.

Gu (1998) used a single fuzzy comprehensive evaluation method to select urban water resources system planning alternatives, which was the first time in China to study urban water resources planning problems with multi-criteria method. Chen (1998) adopted a two-tiered Zadeh fuzzy pattern recognition model in the priority selection of pumped-storage power station. Wang (2000) introduced the modern system theory and method into large-scale water resources planning and management decision-making analysis; they use fuzzy comprehensive evaluation methods and artificial neural network to optimize decision-making. Chen et al. (2002) described the application of multi-level, multi-objective group decision-making theory in the optimization and planning of large-scale water resources systems. Yu and Wang (2004) proposed a multi-person multi-objective decision-making model for the problems of flood control operation in Songhua River Catchment in China.

Bender and Simonovic (2000) described a fuzzy compromise approach to water resources systems planning under uncertainty. Pavlou and Stansbury (1998) applied a formal analysis of the trade-off between environmental risk reduction and cost to contaminated sediment disposal. The approach entails the application of decision analytical tools to build consensus among stakeholders and the general public in selecting a preferred alternative. Stansbury et al. (1999) developed a risk-cost analysis methodology that evaluates uncertainty directly in the decision framework. That is a multi-criteria decision-making method which incorporates uncertainties using fuzzy set theory and composite programming. Liu and Wirtz (2007) integrated a second order fuzzy comprehensive evaluation method and consensus facilitating techniques into computerized group decision support system to minimize pollution effects of oil spill risk in coastal areas. Lahdelma et al. (1998) used MCDA methods in environmental planning and management to choose alternatives with non-commensurate criteria and different stakeholders with conflicting preferences. Apostolakis et al. (2001) developed a single framework integrating risk assessment and decision analysis methods that use MAUT and risk assessment techniques for evaluating, ranking and selecting preferred remediation alternatives at a contaminated site. Based on multi-attribute utility theory and a graph theory-based network analysis algorithm, Michaud and Apostolakis (2006) presented a scenario-based methodology for the ranking of the elements of a water-supply network according to their value to the network's owner. Accorsi et al. (1999a and b) presented an approach based on the AHP and accounts for linguistic imprecision in the stakeholder assessments to construct the utility functions that decision analysis requires the preferences input of each stakeholder. Tony Prato (2003) used multiple attribute evaluation to score and rank five management alternatives for the Missouri River system which are characterized by 10 attributes. Bonano et al. (2000) developed and demonstrated a single framework integrating risk assessment and decision analysis methods relying on stakeholder inputs for evaluating, ranking and selecting preferred remediation alternatives at a contaminated site. Taking into account public participation in environmental decisions, Gregory et al. (2001) articulated a decision aiding method based on constructive, multi-attribute techniques and value-focused thinking to seek consensus among members of a multi-party stakeholder group. Hamalainen et al. (2000) used multi-attribute utility theory tool to analyze risks in nuclear emergency management and planning which should deal with conflicting objectives, different parties involved, and uncertainties. Brito and de Almeida (2009) presented a decision model for risk assessment and for risk ranking of sections of natural gas pipelines based on multi-attribute utility theory. By using decision analysis concepts, this model quantitatively incorporates the DMs' preferences and behavior regarding risk within clear and consistent risk measurements. Abrishamchi et al. (2005) presented an attempt to put into practice the multi-criteria decision-making technique of compromise programming for a real urban water management case study in the city of Zahidan in Iran. Compromise programming is applied to aiding decision makers in selecting the best possible alternatives for distribution of both available and the transmitted water in the city.

In the next section, basic information on FMCGDM model is presented, following by a step-by-step procedure on the development of a risk-based fuzzy multi-criteria group decision-making model in aiding decision-making process.

2 Outline of the FMCGDM Methodology

2.1 The Advantages of FMCGDA for Ecological Risk Management

Ad-hoc decision making is what we naturally do when faced with an issue, especially when there is not time or other incentive to take a more structured approach. Exactly what people do “ad-hoc” depends on the problem; the person; social context and time available. Ad-hoc (also experimental) decision making might cause the selection of a wrong alternative, such as what happened in the risk decision-making of Songhua River water pollution controlling scenario. Ecological risk assessment and management always face with multiple risk sources and stressors, different receptors and endpoints, divergent administrative organization and management goals, diverse assessment criteria and limiting factors. On the issue of formulating ecological risk management options, experiential decision-making mainly rely on DMs’ own knowledge systems and it isn’t an integral part to consult from experts before a decision is made. Traditional MCDA is then applied in the field of ecological risk management to take into account and aggregate decision makers’ judgments on the management alternatives according to different criteria. With the development of applying MCDA methodology into ecological risk management, researchers find out that environmental risk decision-making process is subjected to scarcity of data and lack of knowledge, which necessitates making decisions under ignorance (Harwell et al. 1986). Even if the data are available, criteria quite often contain linguistic definitions involving human judgment and subjectivity, which introduce uncertainties in the decision-making process (Tefamariam and Sadiq 2006). Hence, risk management is a process of informed decision- making under vagueness type uncertainty and partial ignorance. This can be resolved by proposing fuzzy theory in the traditional MCDA. Meanwhile, Groups are seen as superior to individuals as decision-making entities because groups can represent a larger and more diverse set of perspectives and with objective criteria, the use of group decision-making are wildly used in option generation, problem solving, or creative processes of an exploratory nature (Wetmore and Summers 2003).

The FMCGDM method used in this paper can be seen as an integral part of a computerized group decision support system to aid managers to make a reasonable decision. This study focuses on the multiple criteria analysis with an integration of fuzzy set theory and group decision making method. The integration achieves the collective benefits of the three frameworks.

2.2 Rationale of FMCGDM Model

Assume that there are n decision-makers D_k ($k = 1, 2, \dots, n$), they are required to evaluate m possible risk management alternatives A_i ($i = 1, 2, \dots, m$). Every decision-maker will independently evaluate all the alternatives, which are characterized in terms of a criteria set C_j ($j = 1, 2, \dots, l$) chosen by each DM himself. $w_1, \dots, w_j, \dots, w_L$ are the expressions of the importance of each criterion (weights), while $W_1, W_2, \dots, W_n$ are the expressions of the relative importance (weights) of DMs in the decision making team. The judgment matrix proposed by different DMs is expressed as R_k ($k = 1, 2, \dots, n$). Different DMs can choose the same or different criteria and weights set to make their independent evaluations. The forms of expression of attributes of a

criterion can be precise (e.g. real number, interval number) or imprecise (e.g. lexical expression, qualitative judgment). All imprecise attributes and their corresponding weights are expressed as triangular fuzzy numbers in this paper. $\tilde{X}$ is the set of all fuzzy numbers.

For a given set of m alternatives and l criteria, there exists a data matrix that contains the decision judgment matrix $\tilde{P}^k = (\tilde{P}_{ij}^k)_{m \times l}$, where i represents the row containing the management alternative, and j represents the column containing the decision criterion. The superscript k represents this judgment matrix is made by the k th decision maker $D_k (i = 1, 2, \dots, m; j = 1, 2, \dots, l; k = 1, 2, \dots, n)$. A matrix $\tilde{\lambda}^k$ is then set up by normalizing each column in $\tilde{P}^k$ between 0 and 1, $\tilde{\lambda}_{ij}^k$ is the expression of normalized $\tilde{P}_{ij}^k$, which can be regarded as the satisfaction degree of decision maker D_k to alternative A_i , based on his judgment about the importance of each effective criterion C_j against alternative A_i . The judgment matrix can also be transformed by assigning weights to the decision criteria. $\tilde{w}_k = (\tilde{w}_{1k}, \tilde{w}_{2k}, \dots, \tilde{w}_{jk}, \dots, \tilde{w}_{lk})$ is a weighting vector, in which element $\tilde{w}_{jk}$ represents the weight for C_j proposed by decision maker D_k , which can be expressed quantitatively or qualitatively; $\tilde{W}_k$ is the measurement of credibility or relative importance (weight) of D_k in the decision makers team. Let fuzzy set $\tilde{B}$ represent the attributes set of an ideal optimal alternative, then, each alternative A_i will have a membership grade $\mu_{\tilde{B}}(A_i)$, which represents the possibility of alternative A_i being chosen as the best option. Its mathematical expression is defined as follows,

$$\tilde{B} = \{(A_i, \mu_{\tilde{B}}(A_i) | A_i \in A)\} \quad (1)$$

The degree of dominance of one alternative over another to achieve risk management goals and objectives is measured by calculating its membership degree. The alternative which owns the highest degree of membership will be chosen as the best option, expressed as $\max_{A_i \in A} \mu_{\tilde{B}}(A_i)$.

2.3 Procedures

Based the universal concept model of MCDA: $F(A_i) = w \otimes x_{ij}$, It is easy to find out that no matter how complex a multi-attribute decision-making problem is, it can be simply divided into four modules, including: (1) weight module w ; (2) performance matrix module x_{ij} ; (3) aggregation arithmetic operator module $\otimes$ and (4) priority ranking approach module $F(A_i)$. In order to describe these four modules more clearly, this paper proposes a two-stage, twelve-step FMCGDM model. A brief procedure scheme is outlined in Fig. 1. A step-by-step description of the methodology is presented in the next sections and the first stage proceeded along six main steps is summarized as following.

2.3.1 Stage 1: Basic Steps

- Step 1: Construct fuzzy decision judgment matrix, that is, to design the alternatives set and evaluation criteria set, according to the actual situation, determine the actual index values of criteria and establish initialized fuzzy performance matrix.
- Step 2: Design the fuzzy number normalization method, and then calculate standardized fuzzy decision matrix.

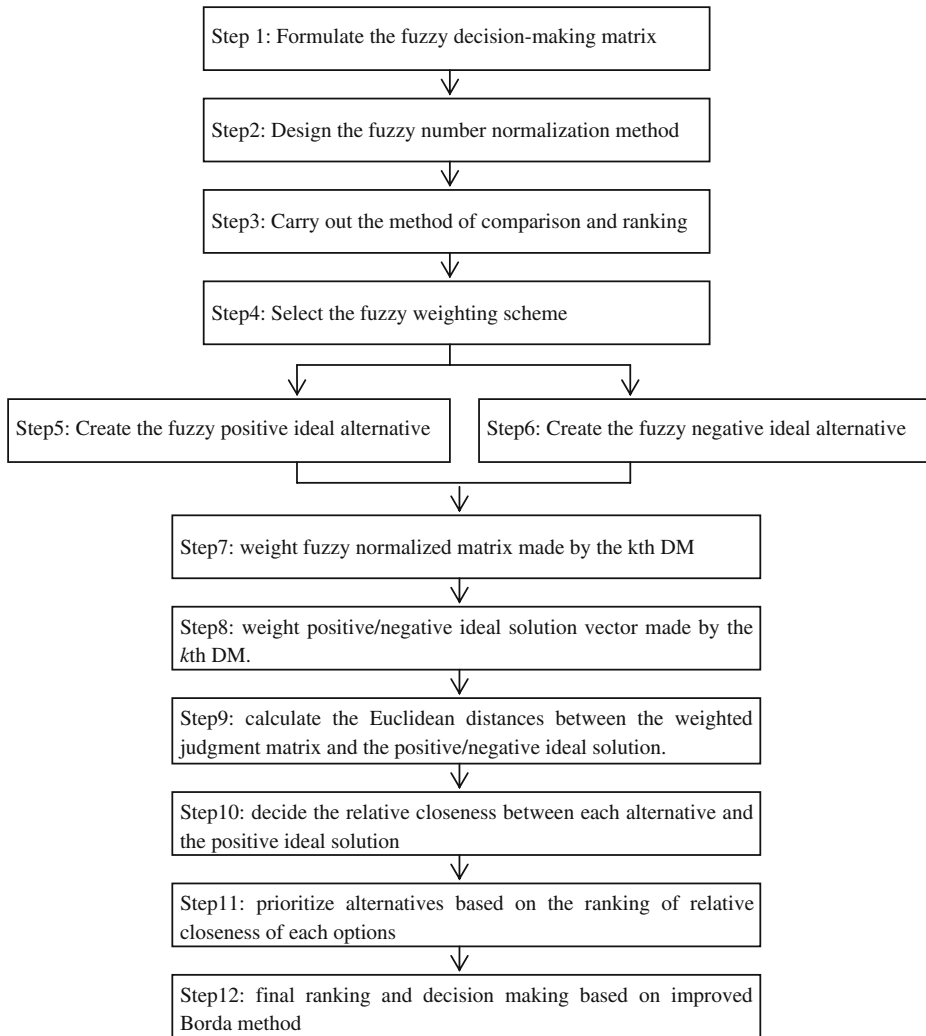


Fig. 1 A 12-step flowchart of the proposed FMCGDM methodology

Step 3: Carry out the method of comparison and ranking. It can be used as the basis for determining the distance between each scheme and the corresponding positive/negative ideal solution. Then, the relative closeness calculation method can be defined with respect to the distances.

Step 4: Select the fuzzy weighting scheme. It is difficult to ask stakeholders or DMs coming from different backgrounds to assign the same value for each criterion and to show the same preference for each alternative. Thus in this study, DMs are supposed to present their preferences for criteria lexically or quantitatively, which means, weights may be precise or vague.

Step 5: Let $\tilde{I}_k^+$ ($k = 1$ to n) represents the k th DM's corresponding fuzzy positive ideal alternative. The final vector of the positive ideal alternative set can be described as,

$$\tilde{I}_k^+ = (\tilde{I}_{1k}^+, \tilde{I}_{2k}^+, \dots, \tilde{I}_{jk}^+, \dots, \tilde{I}_{Lk}^+) \quad (2)$$

Where $\tilde{I}_{jk}^+$ ($j = 1$ to L) is the positive ideal point of the j th criterion for specific alternative, defined by:

For benefit-type indexes

$$\tilde{I}_{jk}^+ = \max_i \{\tilde{\lambda}_{ij}^k\}$$

For cost-type indexes

$$\tilde{I}_{jk}^+ = \min_i \{\tilde{\lambda}_{ij}^k\}$$

Step 6: Let $\tilde{I}_k^-$ ($k = 1$ to n) represents the k th DM's corresponding fuzzy negative ideal alternative. The final vector of the negative ideal alternative set can be described as,

$$\tilde{I}_k^- = (\tilde{I}_{1k}^-, \tilde{I}_{2k}^-, \dots, \tilde{I}_{jk}^-, \dots, \tilde{I}_{Lk}^-) \quad (3)$$

Where $\tilde{I}_{jk}^-$ ($j = 1$ to L) is the negative ideal point of the j th criterion for specific alternative, defined by:

For benefit-type indexes

$$\tilde{I}_{jk}^- = \min_i \{\tilde{\lambda}_{ij}^k\}$$

For cost-type indexes

$$\tilde{I}_{jk}^- = \max_i \{\tilde{\lambda}_{ij}^k\}$$

2.3.2 Stage 2: Fuzzy Group Decision-making Model Based on Weighted Borda Scoring Method

For the ease of narration, the following assumptions are predefined in this section: DMs use the same fuzzy TOPSIS model, select the same criteria set for each of the alternatives and the weight determining methods for each of the criteria are also the same. While, the DMs' preference for each alternative (w_j , $j = 1$ to L) is definitely different, the influence and importance of each of the DMs (W_k , $k = 1, \dots, n$) in the decision group are all different too.

Step 7: Weight the fuzzy normalized matrix made by the k th DM. its element can be determined:

$$\tilde{r}_{ij}^k = \tilde{w}_{jk} \odot \tilde{\lambda}_{ij}^k, \quad \forall i, j, \quad k = 1, \dots, n \quad (4)$$

Step 8: Weight the positive/negative ideal solution vector proposed by the k th DM. its element can be obtained through the following formula:

$$\tilde{R}_{jk}^+ = w_{jk} \tilde{I}_{jk}^+ \quad (5)$$

$$\tilde{R}_{jk}^- = w_{jk} \tilde{I}_{jk}^- \quad (6)$$

Step 9: Calculate the Euclidean distances between the i th alternative of the k th DM $\tilde{r}_{ij}^k$ and the positive/negative ideal solution $\tilde{R}_{jk}^+ / \tilde{R}_{jk}^-$.

$$F_{ik}^+ = \left\{ \sum_{j=1}^L \left[d\left(\tilde{r}_{ij}^k, \tilde{R}_{jk}^+\right) \right]^2 \right\}^{\frac{1}{2}}, \quad i=1,2,\dots,m \quad (7)$$

$$F_{ik}^- = \left\{ \sum_{j=1}^L \left[d\left(\tilde{r}_{ij}^k, \tilde{R}_{jk}^-\right) \right]^2 \right\}^{\frac{1}{2}}, \quad i=1,2,\dots,m \quad (8)$$

where,

$$d\left(\tilde{r}_{ij}^k, \tilde{R}_{jk}^+\right) = \beta_k d\left(\tilde{r}_{ijL}^k, \tilde{R}_{ijkL}^+\right) + (1 - \beta_k) d\left(\tilde{r}_{ijR}^k, \tilde{R}_{ijkR}^+\right) \quad (9)$$

$$d\left(\tilde{r}_{ij}^k, \tilde{R}_{jk}^-\right) = \beta_k d\left(\tilde{r}_{ijL}^k, \tilde{R}_{ijkL}^-\right) + (1 - \beta_k) d\left(\tilde{r}_{ijR}^k, \tilde{R}_{ijkR}^-\right) \quad (10)$$

Where β_k is the risk preference index of decision maker D_k when facing with the potential vagueness type uncertainties inhered in the alternatives (Kong 2008). $\beta_k = 0.5$ demonstrates D_k as a risk neutral, $\beta_k > 0.5$ indicates that decision maker D_k is a risk averter, $\beta_k < 0.5$ means that the k th DM is a risk lover.

Step 10: Decide the relative closeness between each alternative and the positive ideal solution.

$$RC_{ik} = \frac{F_{ik}^-}{F_{ik}^- + F_{ik}^+} \quad (11)$$

Step 11: Rank RC_{ik} according to descending order, which means the former element is better than the latter.

Step 12: Final ranking and decision making based on improved Borda method.

Borda scoring method can be expressed as: let m be the number of ecological risk management alternatives, and n be the number of DMs, each DM can evaluate and rank all the alternatives according to the aforementioned procedures (from step 1 to step 11). Then, the alternative at the end of the ranking sequence will be scored 0 points, the penultimate alternative will be assigned 1 point, the antepenult one will be given 2 points, and so on. All the scores will be summed, which is called Borda score of the alternative. The higher the score is, the better the alternative is.

Table 1 Ultimate scoring procedure with Borda scoring method

Alternatives	DMs								Total score
	D ₁		D ₂		...		D _n		
	Borda	Weight	Borda	Weight	Borda	Weight	Borda	Weight	
	score		score		score		score		
A ₁	b ₁₁	W ₁	b ₁₂	W ₂	...	W...	b _{1n}	W _n	$S_i = \sum_{k=1}^n W_k b_{ik}$
A ₂	b ₂₁		b ₂₂		...		b _{2n}		
...	...		...		...		...		
A _m	b _{m1}		b _{m2}				b _{mn}		
Ranking	In terms of the S _i , ranking every alternative in accordance with the descending order, the higher the score is, the better the alternative.								

As mentioned above, in real-life decision-makers team, the experience, knowledge background and influence of each DM are all different, this means not all the decision makers are equally important. Hence, a weighting vector $W = (W_1, W_2, \dots, W_k, \dots, W_n)$ is used to adjust the importance of each decision maker. It can be characterized as fixed number, fuzzy number, lexical description or fuzzy interval number et al. If they are defined as fixed numbers, the following restricted condition should be satisfied:

$$\sum_{k=1}^n W_k = 1 \quad (12)$$

Where $0 \leq W_k \leq 1$.

The product of initial Borda score and W_k of each alternative will be re-ranked in descending order; the alternative with the highest score will be regarded as the best risk management program. The calculation process is illustrated in Table 1.

3 Case Study

As far as watershed ecological risk management decision-making plan be concerned, which involves local/province/state environmental protection department, water conservancy department, ministry of land and resources, finance sector and other responsible authorities, also including managers, local residents, business owners, environmental risk assessors and other stakeholders, a sound and feasible decision should not only be made on the viewpoint of only one decision maker or one department without caring about the attitudes of other decision makers and stakeholders, especially those who will really live with the potential risks. Because of different evaluation criteria set and risk preferences, different DMs will have competing or contradictory understanding on ecological risks suffered by a watershed. So, watershed ecological risk management can be interpreted as a multi-criteria group decision-making problem involving a variety of stakeholder interests and complex natural environments.

In this paper, the China Yangtze River Three Gorges reservoir area risk assessment and management process is chosen as a study case to analyze the application of such a FMAGDM method in watershed ecological risk management. It is a simplified semi-scenario analysis case developed for the purpose of demonstration

in accordance with the real situation of the Three Gorges reservoir area and real stakeholders.

The Three Gorges Reservoir Project is the largest water control project of China at the present time, its total watershed area reaches 54,000 km². By the end of year 2007, the total population in the reservoir area is more than 20 million, with a GDP of more than 300 billion CNY. Three Gorges reservoir area is one of the most important ecologically fragile areas among the upper reaches of Yangtze River. The related decision-making, such as choosing the best scheme of maintaining local ecological health, planning the ecological risk early warning and contingency management system and proposing the counter measures, involves a vast variety of stakeholders, uncertain factors, evaluation criteria and multiple alternatives. Therefore, its ecological risk prevention and controlling management process can be addressed by a FMAGDM method mentioned above.

For such a complex process as Three Gorges reservoir catchment ecological risk management, it is difficult for us to make an absolutely optimal operation choice, but it is possible to find a relatively optimal alternative from the given feasible alternatives. Before choosing the best watershed risk management option, the highest competent authority first set up an expert panel, involving various experts and researchers from twelve science research institutes and universities, including the authors of this paper, to make comprehensive ecological security investigation and assessment on the reservoir area, in 2007. The expert team set up a hierarchical structure to comprehensively investigate and evaluate the ecological situations of this catchment. Four criteria, including the loss of ecosystem services in the reservoir area, watershed socio-economic losses, ecological catastrophes, stability of the aquatic ecosystem in the reservoir area were chosen as the level-I attributes by the expert term. Then, a either two or three level integrated evaluation system was established and the final aggregative risk for this specific watershed was proposed. At the end of 2008, four feasible ecological risk management alternatives were put forward by the expert panel for the final decision-making. Because of the confidentiality requirement, this paper will not represent the specifics of the four options, but discuss the ecological management decision making method itself.

3.1 Construct the Initial Judgment Matrix

Assume that Environmental Management Department (D_1), Ministry of Water Resources (MWR, D_2), National Development and Reform Commission (NDRC, D_3), delegation of local public opinion (D_4) be the four DMs D_k ($k = 1, 2, 3, 4$), each DM will judge the effectiveness of four alternatives A_i ($i = 1, 2, 3, 4$) respectively. Five criteria C_j ($j = 1, 2, 3, 4, 5$), including satisfying the ecosystem services (C_1), enhancement of socio-economic development in the reservoir area (C_2), prevention of ecological disaster (C_3), maintaining the stability of aquatic ecosystem (C_4) and the economic feasibility (investment scale) of the alternatives (C_5) are proposed as the criteria set. Assume that the priority of each criterion, the DMs' preference for each alternative and the influence of every DM in the decision-making team are all different. The judgments made by D_k ($k = 1, 2, 3, 4$) are expressed with linguistic variables by a nine-level fuzzy qualitative scale which is defined by triangular fuzzy numbers (TFNs). The subjective weights in the last row of the following four tables (Tables 2, 3, 4, 5) were assigned by the expert panel using Delphi method.

Table 2 Judgment matrix R_1 proposed by DM_1 (D_1)

EP extremely poor, QP quite poor, MP mildly poor, MG mildly good, QG quite good, EG extremely good, QL quite poor

Alternatives	Criteria				
	C_1	C_2	C_3	C_4	C_5
A_1	QG	MP	MG	MG	QG
A_2	MP	QG	Ok	MP	QP
A_3	Ok	Ok	QG	MG	MP
A_4	MG	Ok	MG	QG	Ok
Weight	0.30	0.10	0.25	0.20	0.15

Table 3 Judgment matrix R_2 proposed by DM_2 (D_2)

Alternatives	Criteria				
	C_1	C_2	C_3	C_4	C_5
A_1	MG	Ok	Good	MG	Ok
A_2	Poor	QG	MP	Poor	Poor
A_3	MG	Ok	QG	Good	Poor
A_4	Good	MP	QG	EG	MP
Weight	0.25	0.20	0.30	0.15	0.10

Table 4 Judgment matrix R_3 proposed by DM_3 (D_3)

Alternative	Criteria				
	C_1	C_2	C_3	C_4	C_5
A_1	Ok	Poor	Ok	MG	Good
A_2	MP	EG	QG	MG	Poor
A_3	Ok	Ok	Good	Good	QP
A_4	MG	MP	MG	EG	Ok
Weight	0.15	0.25	0.20	0.10	0.30

Table 5 Judgment matrix R_4 proposed by DM_3 (D_4)

Alternative	Criteria				
	C_1	C_2	C_3	C_4	C_5
A_1	Good	MP	MG	Ok	Ok
A_2	Ok	Good	Good	Good	QP
A_3	Ok	MG	EG	Ok	EP
A_4	Good	QP	MG	QG	Poor
Weight	0.20	0.30	0.15	0.25	0.10

Table 6 The nine-level ranking system in terms of linguistic classification of grades and its corresponding TFNs

Ranking level	Qualitative explanations for grade of performance	TFNs	Standardized TFNs
1	Extremely good	(14,16,16)	(0.875,1,1)
2	Quite good	(12,14,16)	(0.75,0.875,1)
3	Good	(10,12,14)	(0.625,0.75,0.875)
4	Mildly good	(8,10,12)	(0.5,0.625,0.75)
5	Ok	(6,8,10)	(0.375,0.5,0.625)
6	Mildly poor	(4,6,8)	(0.25,0.375,0.5)
7	Poor	(2,4,6)	(0.125,0.25,0.375)
8	Quite poor	(0,2,4)	(0,0.125,0.25)
9	Extremely poor	(0,0,2)	(0,0,0.125)

3.2 Design Fuzzy Numbers Normalization Method, Set Up Normalized Fuzzy Decision Matrix

According to psychologist GA Miller's research, when judging the performance of a criteria for an alternative, common people can correctly distinguish their differences from five-level scale to nine-level scale. Namely, lexical indicators can be divided into a five-level to nine-level ranking system by which grade and importance of alternative's criteria can be classified. Table 6 describes a nine-level qualitative scaling system for grade of alternatives' criteria. Each qualitative scale describing the vagueness and fuzziness in the lexical connotation is expressed by TFNs; the standardized TFNs are listed in the last column.

Thus, the standardized fuzzy judgment matrix in Tables 2, 3, 4, 5 can be translated into following forms: (Tables 7, 8, 9, 10).

3.3 Design the Comparison and Ranking Method of Fuzzy Numbers

As far as the ranking of fuzzy numbers with regard to fuzzy MCDM problems is concerned, usually the distance between a fuzzy number and the reference basis will be calculated. In this paper, we take the fuzzy distances between fuzzy numbers and the negative-ideal point as the ranking indices and also take every DM's preference for uncertainties contained in every alternative into consideration. The process can be expressed as follows

$$F_i = d(\tilde{x}_i, \tilde{m}^-) = \beta d(\tilde{x}_{iL}, \tilde{m}_L^-) + (1 - \beta) d(\tilde{x}_{iR}, \tilde{m}_R^-) \quad (13a)$$

If the distance between the positive and negative ideal point $d(\tilde{m}^+, \tilde{m}^-)$ is not equal to 0, the above formula can be transformed into Eq. 13b, which can be regarded as a normalization of Eq. 13a.

$$F_i = \frac{d(\tilde{x}_i, \tilde{m}^-)}{d(\tilde{m}^+, \tilde{m}^-)} = \frac{\beta d(\tilde{x}_{iL}, \tilde{m}_L^-) + (1 - \beta) d(\tilde{x}_{iR}, \tilde{m}_R^-)}{\beta d(\tilde{m}_L^+, \tilde{m}_L^-) + (1 - \beta) d(\tilde{m}_R^+, \tilde{m}_R^-)} \quad (13b)$$

Where β_k is the risk preference index of decision maker D_k when facing with the potential vagueness type uncertainties inhered in the alternatives. $\beta_k = 0.5$ demonstrates D_k as a risk neutral, $\beta_k > 0.5$ indicates that decision maker D_k is a risk averter, $\beta_k < 0.5$ means that the k th DM is a risk lover.

Where $(\tilde{m}_L^+, \tilde{m}_R^+)$ and $(\tilde{m}_L^-, \tilde{m}_R^-)$ are left fuzzy set and right fuzzy set of set of positive and negative ideal point, d is the hamming distance, β is the acceptable level of the DM with regard to the potential uncertainties (risk preference) inhered in the alternatives. $\beta = 0.5$ represents the DM as a risk neutral, $\beta < 0.5$ means the DM as a risk lover, $\beta > 0.5$ indicate the DM as a stakeholder who is unwilling to take uncertainty (risk averter). The computing formula of hamming distance can be expressed as,

$$d_H(A, B) = \int |\mu_A(x) - \mu_B(x)| dx \quad (14)$$

Table 7 Standardized TFNs judgment matrix R'_1 proposed by DM_1 (D_1)

Alternatives	Criteria				
	C_1	C_2	C_3	C_4	C_5
A_1	(0.75,0.875,1)	(0.25,0.375,0.5)	(0.5,0.625,0.75)	(0.5,0.625,0.75)	(0.5,0.625,0.75)
A_2	(0.25,0.375,0.5)	(0.75,0.875,1)	(0.375,0.5,0.625)	(0.25,0.375,0.5)	(0.0,125,0.25)
A_3	(0.375,0.5,0.625)	(0.375,0.5,0.625)	(0.75,0.875,1)	(0.5,0.625,0.75)	(0.25,0.375,0.5)
A_4	(0.5,0.625,0.75)	(0.375,0.5,0.625)	(0.5,0.625,0.75)	(0.75,0.875,1)	(0.375,0.5,0.625)
Weight	0.30	0.10	0.25	0.20	0.15

Table 8 Standardized TFNs judgment matrix R'_2 proposed by DM_2 (D_2)

Alternatives	Criteria				
	C_1	C_2	C_3	C_4	C_5
A_1	(0.5,0.625,0.75)	(0.375,0.5,0.625)	(0.625,0.75,0.875)	(0.5,0.625,0.75)	(0.375,0.5,0.625)
A_2	(0.125,0.25,0.375)	(0.75,0.875,1)	(0.25,0.375,0.5)	(0.125,0.25,0.375)	(0.125,0.25,0.375)
A_3	(0.5,0.625,0.75)	(0.375,0.5,0.625)	(0.75,0.875,1)	(0.625,0.75,0.875)	(0.125,0.25,0.375)
A_4	(0.625,0.75,0.875)	(0.25,0.375,0.5)	(0.75,0.875,1)	(0.875,1,1)	(0.25,0.375,0.5)
Weight	0.25	0.20	0.30	0.15	0.10

Table 9 Standardized TFNs judgment matrix R'_3 proposed by DM_3 (D_3)

Alternatives	Criteria				
	C_1	C_2	C_3	C_4	C_5
A_1	(0.375,0.5,0.625)	(0.125,0.25,0.375)	(0.375,0.5,0.625)	(0.5,0.625,0.75)	(0.625,0.75,0.875)
A_2	(0.25,0.375,0.5)	(0.875,1,1)	(0.75,0.875,1)	(0.5,0.625,0.75)	(0.125,0.25,0.375)
A_3	(0.375,0.5,0.625)	(0.375,0.5,0.625)	(0.625,0.75,0.875)	(0.625,0.75,0.875)	(0.0,125,0.25)
A_4	(0.5,0.625,0.75)	(0.25,0.375,0.5)	(0.5,0.625,0.75)	(0.875,1,1)	(0.375,0.5,0.625)
Weight	0.15	0.25	0.20	0.10	0.30

Table 10 Standardized TFNs judgment matrix R'_4 proposed by DM_4 (D_4)

Alternatives	Criteria				
	C_1	C_2	C_3	C_4	C_5
A_1	(0.625,0.75,0.875)	(0.25,0.375,0.5)	(0.5,0.625,0.75)	(0.375,0.5,0.625)	(0.375,0.5,0.625)
A_2	(0.375,0.5,0.625)	(0.625,0.75,0.875)	(0.625,0.75,0.875)	(0.625,0.75,0.875)	(0.0,125,0.25)
A_3	(0.375,0.5,0.625)	(0.5,0.625,0.75)	(0.875,1,1)	(0.375,0.5,0.625)	(0.0,125)
A_4	(0.625,0.75,0.875)	(0.0,125,0.25)	(0.5,0.625,0.75)	(0.75,0.875,1)	(0.125,0.25,0.375)
Weight	0.20	0.30	0.15	0.25	0.10

3.4 Weighting Schemes

3.4.1 Methods of Weighting

Various techniques are used to compute the final fuzzy weights. Normally, weights of the experts and attributes can be classified into two types: subjective weights and objective weights. The former reflect the subjective preference of decision-maker, the latter demonstrate the natural property of attributes. Taking the complex conditions of determining the relative importance of criteria/attributes in the judgment matrix into consideration, this paper proposes an integrated approach to determine criteria weights in the multiple criteria decision analysis (MCDA) problems. With a weight tradeoff coefficient, the approach makes use of the subjective information provided by a decision maker and the objective information of the proposed criteria to form hybrid weights. Thus the resultant attribute weights and rankings of alternatives reflect both the subjective considerations of a decision maker (DM) and the objective information.

So-called hybrid weighting method is to combine multiple weights firstly determined with different methods together in accordance with pre-set rules and form the final weights set. Firstly, according to characteristics of the ecological risk decision-making program, appropriate single subjective/objective methods are selected to calculate initial weights respectively. Secondly, the operator for initial weights combination is set to come up with the optimal rule of combination. Assume that $W_i (i = 1, 2, \dots, n)$ are subjective weights proposed for the given criteria, and $W_j (j = 1, 2, \dots, n)$ are corresponding objective weights, and then the hybrid weight formula can be expressed as

$$W_g = \theta W_i \oplus (1 - \theta) W_j \quad (15)$$

Where θ is the tradeoff coefficient of decision-maker weight, $0 \leq \theta \leq 1$, reflecting the preferences for subjective weight and objective weight against different DMs. $\oplus$ is the aggregation operator. Assume that every DM may employ the same weighting methods in the group decision-making process, two objective weighting methods and two subjective weighting methods are denoted as following.

3.4.2 Fuzzy Subjective Weighting Method

Common fuzzy subjective weighting methods include the fuzzy eigenvector weighting method, fuzzy judging matrix weighting method, fuzzy complementary pairwise comparison weighting method, etc. Delphi method is a familiar subjective weighting method. Apart from Delphi method, we take fuzzy complementary pairwise comparison weighting method (FCPCW) as another way to sort the relative importance of different criteria in a judgment matrix. The basic process is to get the fuzzy judgment matrix $A = [a_{ij}]_{n \times n}$ through the comparison of importance of every two attributes. Where, $a_{ij} = \frac{w_i}{w_j}$ is the measure of importance degree between the i th criterion and the j th criterion.

This paper takes the relative most important criterion in the criteria set as the comparison basis, then, each criterion in the criteria set is compared to this basis. Taking DM D_1 as an example, the pre-set Delphi subjective weights in Table 2 can be translated into a fuzzy comparison vector ($\tilde{V}$) of the decision-making indicators

matrix, which is listed in Table 11. The scales used to transform fuzzy lexical judgments into fuzzy numbers in are tabulated in Table 12.

In theory, based on the fuzzy comparison vector $\tilde{V}$, the total fuzzy judgment matrix can be determined by the characteristics of consistency discriminant matrix. However, this would be a very complicated mathematical process. So practically, we decompose the fuzzy comparison vector ($\tilde{V}$) into several judgment submatrix to simplify the calculation process. The specific method is to break down $\tilde{V}$ into $1 + 2^{n-1}$ judgment submatrix, where the number l represents the sub-vector composed of the TFNs medians of each element in the fuzzy comparison vector $\tilde{V}_1 = (a_{11}^n, \dots, a_{1n}^n)$. The rest 2^{n-1} sub-vectors are derived from the permutation and combination of upper boundary value and lower boundary value of the elements $(1, 2, \dots, n)$ of $\tilde{V}$, denoted as $V_1, V_2, \dots, V_k$ ($k = 2^{n-1}$).

Based on the $1 + 2^{n-1}$ sub-vectors, using the following characteristic Eq. 16 of consistency judgment matrix of the complementary pairwise comparison method, we can get the corresponding judgment submatrix, expressed as $A_1, A_2, \dots, A_k$ ($k = 2^{n-1}$).

$$a_{ii} = 1; a_{ij} = 1/a_{ji}; a_{ij} = a_{ik}/a_{jk} \quad (16)$$

In this case, there are five criteria ($n = 5$). So, we can get 17 sub-vectors as follows

$$\begin{aligned} V_m &= (1, 3, 5, 7, 9), V_1 = (1, 2, 4, 6, 8), V_2 = (1, 2, 4, 6, 9), V_3 = (1, 2, 4, 8, 8), \\ V_4 &= (1, 2, 4, 8, 9), V_5 = (1, 2, 6, 6, 8), V_6 = (1, 2, 6, 6, 9), V_7 = (1, 2, 6, 8, 8), \\ V_8 &= (1, 2, 6, 8, 9), V_9 = (1, 4, 4, 6, 8), V_{10} = (1, 4, 4, 6, 9), V_{11} = (1, 4, 4, 8, 8), \\ V_{12} &= (1, 4, 4, 8, 9), V_{13} = (1, 4, 6, 6, 8), V_{14} = (1, 4, 6, 6, 9), V_{15} = (1, 4, 6, 8, 8), \\ V_{16} &= (1, 4, 6, 8, 9) \end{aligned}$$

With the corresponding judgment submatrix (see Appendix) and MATLAB2007 R, 17 weight elements of the five criteria are calculated and list in Table 13.

Based on the definition of TFN, which is represented by three points (a, b, c) on the universe of discourse representing the minimum, most likely, and maximum values, respectively, from Table 17, we can get the fuzzy combination weight vector $\tilde{W} = (\tilde{w}_1, \tilde{w}_2, \tilde{w}_3, \tilde{w}_4, \tilde{w}_5)$, where,

$$\begin{aligned} \tilde{w}_1 &= (0.490, 0.559, 0.605); \\ \tilde{w}_2 &= (0.055, 0.062, 0.075); \\ \tilde{w}_3 &= (0.140, 0.187, 0.263); \\ \tilde{w}_4 &= (0.085, 0.112, 0.144); \\ \tilde{w}_5 &= (0.063, 0.080, 0.098). \end{aligned}$$

Table 11 Fuzzy comparison results the of Delphi subjective weights proposed by D₁

	C ₁	C ₃	C ₄	C ₅	C ₂
Result of comparison between C ₁ and C _i	Identical	Slightly important	Important	Very important	Extremely important
Weight proportion of C ₁ compared to C _i	Identical	Slightly high	High	Very high	Extremely high
Corresponding TFNs	(1,1,1)	(2,3,4)	(4,5,6)	(6,7,8)	(8,9,9)

Table 12 Overview of fuzzy numbers used for scaling lexical judgments

Importance degree	Lesser important	Identical	Slightly important	Important	Very important	Extremely important
Corresponding TFNs	(0.9,1,1)	(1,1,1)	(2,3,4)	(4,5,6)	(6,7,8)	(8,9,9)
Corresponding trapezoidal fuzzy numbers	(0.8,0.9,1,1,1.2)	(1,1,1,1,1)	(1.5,2.5,3.5,4.5)	(3.5,4.5,5.5,6.5)	(5.5,6.5,7.5,8.5)	(7.5,8.5,9.5,9.9)
Corresponding fuzzy interval numbers	(0.9,1.2)	(1,1)	(2,4)	(4,6)	(6,8)	(8,9)

Table 13 17 weight values of the five criteria

	C ₁	C ₃	C ₄	C ₅	C ₂
W _m	0.559	0.187	0.112	0.080	0.062
W ₁	0.490	0.245	0.122	0.082	0.061
W ₂	0.493	0.247	0.123	0.082	0.055
W ₃	0.500	0.250	0.125	0.063	0.063
W ₄	0.504	0.252	0.126	0.063	0.056
W ₅	0.511	0.255	0.085	0.085	0.064
W ₆	0.514	0.257	0.086	0.086	0.057
W ₇	0.522	0.261	0.087	0.065	0.065
W ₈	0.526	0.263	0.088	0.066	0.058
W ₉	0.558	0.140	0.140	0.093	0.070
w ₁₀	0.563	0.141	0.141	0.094	0.063
w ₁₁	0.571	0.143	0.143	0.071	0.071
w ₁₂	0.576	0.144	0.144	0.072	0.064
w ₁₃	0.585	0.146	0.098	0.098	0.073
w ₁₄	0.590	0.148	0.098	0.098	0.066
w ₁₅	0.600	0.150	0.100	0.075	0.075
w ₁₆	0.605	0.151	0.101	0.076	0.067

3.4.3 Fuzzy Objective Weighting Method

In this paper, the combination of fuzzy range weight (FRW) and fuzzy entropy weight (FEW) is presented to reflect the fuzzy objective weight for the criterion. Defining the modified hamming distance from x_{ij} to x_j^- ($i = 1, 2, 3, 4; j = 1, 2, 3, 4, 5$) as:

$$F_{ij} = d(x_{ij}, x_j^-) = \beta_k d(x_{ijL}, x_{jL}^-) + (1 - \beta_k) d(x_{ijR}, x_{jR}^-) \quad (17)$$

Where x_{ij} is the element value of the j th criterion for the i th alternative, x_j^- is the negative-ideal point of j th criterion, x_{jL}^- is the left fuzzy number of x_j^- , x_{jR}^- is the right fuzzy number of x_j^- . The shorter the F_{ij} is, the less important the x_{ij} is.

Let Ψ_j be the fuzzy range of the j th criterion, E_j be the fuzzy entropy of the j th criterion,

$$\Psi_j = \max(F_{1j}, F_{2j}, \dots, F_{mj}) - \min(F_{1j}, F_{2j}, \dots, F_{mj}) \quad (18)$$

$$E_j = - \sum_{i=1}^m (f_{ij} \ln f_{ij}), \text{ where } f_{ij} = \frac{F_{ij}}{\sum_{i=1}^m F_{ij}} \quad (19)$$

The equation of fuzzy range weighting method is:

$$w_j^\Psi = \frac{\Psi_j}{\sum_{j=1}^n \Psi_j} \quad (20)$$

The equation of fuzzy entropy weighting method is:

$$w_j^E = \frac{\ln n - E_j}{\sum_{j=1}^n (\ln n - E_j)} \quad (21)$$

Table 14 F_{ij} , fuzzy range and fuzzy entropy weights for the criteria proposed by DM_1 , $\beta_1 = 0.6$

F_{ij}	C_1	C_2	C_3	C_4	C_5
A_1	0.7375	0.2375	0.4875	0.4875	0.4875
A_2	0.2375	0.7375	0.3625	0.2375	0.0625
A_3	0.3625	0.3625	0.7375	0.4875	0.2375
A_4	0.4875	0.3625	0.4875	0.7375	0.3625
Fuzzy range	0.5	0.5	0.375	0.5	0.425
Fuzzy entropy	1.3052	1.2963	1.3530	1.3173	1.2118
Range weights	0.217	0.217	0.163	0.217	0.185
Entropy weights	0.181	0.201	0.074	0.154	0.390

Taking $DM D_1$ as an example, since all the criteria are standardized in accordance with the 9-level lexical ranking system, the same negative-ideal point can be expressed as a TFN, $x_j^- = (0, 0, 0.125)$. According to Eq. 17, all the F_{ij} are calculated and listed in Table 14.

3.4.4 Methods of Determining the Final Composite Weights

In this case, based on expert judgment method, all four decision-making units, including Environmental Management Department (D_1), Water Department (D_2), National Development and Reform Commission (NDRC, D_3) and delegation of local public stakeholder (D_4) agreed that the importance of subjective weighting method and objective weighting method is different, in Eq. 15, $\theta = 0.4$. Meanwhile, the same preferences (relative importance) are assigned to the sub-objective/subjective weighting methods (W_i, W_j). Because fuzzy weights do not need to be normalized, the ultimate combined weight $w_j^c = (w_{jl}^c, w_{jm}^c, w_{jr}^c)$ for j th criterion can be calculated as follow

$$w_j^c = \prod_{k=1}^t (w_j^k)^{a_k} = \left(\prod_{k=1}^t (w_{jl}^k)^{a_k}, \prod_{k=1}^t (w_{jm}^k)^{a_k}, \prod_{k=1}^t (w_{jr}^k)^{a_k} \right) \quad (22)$$

Where t is the amount of sub-objective/subjective weights, a_k is the preference degree (relative importance) of the k th sub-objective/subjective weight in determining the final combined weight.

Taking D_1 as an example, the weights information is listed in Table 15.

Table 15 Weights information of criteria $\theta = 0.4$

Weighting methods	Subjective weighting methods (0.4)		Objective weighting methods (0.6)	
	Delphi (0.5)	FCPCW (0.5)	FRW (0.5)	FEW (0.5)
Relative importance a_k	0.2	0.2	0.3	0.3
C_1	0.30	(0.49,0.560,0.605)	0.217	0.181
C_2	0.10	(0.055,0.062,0.075)	0.217	0.201
C_3	0.25	(0.085,0.112,0.144)	0.163	0.074
C_4	0.20	(0.063,0.08,0.098)	0.217	0.154
C_5	0.15	(0.140,0.187,0.263)	0.185	0.390

Table 16 Weighted judgment matrix R_1 proposed by D_1

Alternatives	Criteria				
	C_1	C_2	C_3	C_4	C_5
A_1	(0.198,0.237,0.275)	(0.034,0.053,0.073)	(0.086,0.114,0.147)	(0.085,0.112,0.141)	(0.079,0.103,0.129)
A_2	(0.066,0.102,0.138)	(0.103,0.124,0.147)	(0.065,0.091,0.122)	(0.042,0.067,0.094)	(0.0,0.021,0.043)
A_3	(0.099,0.136,0.172)	(0.052,0.071,0.092)	(0.129,0.160,0.196)	(0.085,0.112,0.141)	(0.039,0.062,0.086)
A_4	(0.132,0.169,0.206)	(0.052, 0.071,0.092)	(0.086,0.114,0.147)	(0.127,0.157,0.188)	(0.059,0.082,0.107)
FPIA	(0.198,0.237,0.275)	(0.103,0.124,0.147)	(0.129,0.160,0.196)	(0.127,0.157,0.188)	(0.079,0.103,0.129)
FNIA	(0.066,0.102,0.138)	(0.034,0.053,0.073)	(0.065,0.091,0.122)	(0.042,0.067,0.094)	(0.0,0.021,0.043)

FPIA fuzzy positive ideal alternative; FNIA fuzzy negative ideal alternative

Table 17 Calculation results of relative closeness $\beta_1 = 0.6$

	Distance to FPIA F_i^+	Distance to FNIA F_i^-	RN_i
A ₁	0.095	0.144	0.635
A ₂	0.175	0.050	0.207
A ₃	0.123	0.089	0.432
A ₄	0.097	0.115	0.567

Table 18 Subjective and objective weighting methods used by each DM and the corresponding parameters

DMs	MPs	Objective weighting methods	θ	β_k
	Subjective weighting methods			
D1	Delphi, FCPCW	FEW, FRW	0.40	0.60
D2	Delphi, FEVW	FEW, FDW	0.35	0.70
D3	Delphi, FEVW	FEW, FCVW	0.50	0.40
D4	Delphi, FCPCW	FEW, FSDW	0.60	0.30

MPs methods and processes, *FEVW* fuzzy eigenvector weighting method, *FDW* fuzzy dispersion weighting method, *FCVW* fuzzy coefficient of variation weighting method, *FSDW* fuzzy standard deviation weighting method

Table 19 RN_i ($i = 1, 2, 3, 4$) and the alternatives' corresponding priorities

	D1		D2		D3		D4	
	RN	Priority	RN	Priority	RN	Priority	RN	Priority
A ₁	0.635	1	0.587	3	0.559	1	0.380	3
A ₂	0.207	4	0.226	4	0.370	3	0.560	1
A ₃	0.432	3	0.673	2	0.140	4	0.393	2
A ₄	0.567	2	0.728	1	0.451	2	0.277	4

Table 20 Ultimate scoring process with Borda method

Alternative	DMs								Total score	Priority
	D ₁		D ₂		D ₃		D ₄			
	Borda	Weight	Borda	Weight	Borda	Weight	Borda	Weight		
	score		score		score		score			
A ₁	3	0.3	1	0.2	3	0.2	1	0.3	2.0	1
A ₂	0		0		1		3		1.1	4
A ₃	1		2		0		2		1.3	3
A ₄	2		3		2		0		1.6	2

According to Eq. 22, the final combined weights for every criterion C_j ($j = 1, 2, 3, 4, 5$) for D_1 can be listed as:

$$W_1 = (0.264, 0.271, 0.275); W_2 = (0.138, 0.141, 0.147); W_3 = (0.173, 0.183, 0.196); \\ W_4 = (0.169, 0.179, 0.188); W_5 = (0.157, 0.165, 0.172)$$

3.5 Calculation of the Relative Nearness Degree

1. The determination of $\tilde{I}_k^+$, $\tilde{I}_k^-$ of the k th DM ($k = 1, 2, 3, 4$)

As to D_1 , according to the principles and methods mentioned in Section 2, the fuzzy positive ideal alternative $\tilde{I}_1^+$ and the fuzzy negative ideal alternative $\tilde{I}_1^-$ are derived as following:

$$\tilde{I}_1^+ = ((0.75, 0.875, 1), (0.75, 0.875, 1), (0.75, 0.875, 1), (0.75, 0.875, 1), \\ (0.5, 0.625, 0.75)) \\ \tilde{I}_1^- = ((0.25, 0.375, 0.5), (0.25, 0.375, 0.5), (0.375, 0.5, 0.625), (0.25, 0.375, 0.5), \\ (0, 0.125, 0.25))$$

2. Calculation of weighted judgment matrix

As to D_1 , according to the context of Table 7 and the results of Section 3.4, the weighted judgment matrix can be expressed as Table 16.

3. Relative nearness (RN)

According to Eqs. 8 to 12, the distances of each alternative to ideal alternative and the corresponding relative closeness is denoted in following Table 17.

As to D_1 , by ranking RN_i in descending order, we can get the priorities of the four initial alternatives as:

$$A_1 > A_4 > A_3 > A_2.$$

3.6 Total Results of All Four DMs

In accordance with the methods mentioned above, the RN_i ($i = 1, 2, 3, 4$) with regard to all four DMs and their corresponding intermediate processes and parameters are listed in Tables 18 and 19.

3.7 The Ultimate Decision-Making Process Based on Modified Borda Scoring Method

According to the method mentioned in Section 2, we define the W_k ($k = 1, 2, 3, 4$) as 0.3, 0.2, 0.2 and 0.3, which means that we treat the attitudes of local environmental management authority and representatives of local people more important than that of MWR and NDRC. The calculation results are listed in Table 20.

From the results presented in above table, the priorities and corresponding ranking level of the four alternatives is:

$$A_1 > A_4 > A_3 > A_2$$

Which means alternative A_1 is the best choice for risk managers, while alternative A_2 is the least desired. The priorities of the four alternatives derived from FMCGDM

are almost completely in line with the actual preferred solution of the highest competent authority, which illustrate the decision-making model proposed in this paper is credible and feasible.

4 Conclusions and Discussion

Taking Three Gorges Reservoir area ecological risk management as a case study, this paper established a FMCGDM model based on modified Borda scoring method. It is a two-stage approach. Firstly, all DMs make their own independent decision respectively, the second phase is to integrate all the aforementioned independent conclusions by using their subjective/objective weights and come to the final preferable conclusion.

In the calculating process, language attributes or linguistic variables are transferred to triangular fuzzy numbers by a nine-level fuzzy qualitative scale. Taking different psychological characteristics of different DMs into consideration, a risk preference factor β is added to distinguish different risk-taking tendency. With full respect to the DMs' subjective preferences for every criterion for every alternative and their tendencies toward different weighting methods, a weight operator with a combination of fuzzy subjective weighting methods and fuzzy objective weighting methods is introduced. Based on traditional TOPSIS model, weighted fuzzy relative closeness are proposed to help the DMs to measure the different advantages and disadvantages of various alternatives. Since the importance, influence, professional quality of each decision-maker will all play a big role in the decision-making process, a modified Borda scoring method is introduced to evaluate and rank the four alternatives, and the alternative with the highest score will be regarded as the best solution program.

It is without any doubt that FMCGDM model can be used in environmental planning and management, including ecological risk decision-making. But ecological security maintenance, ecological risk prevention or ecological health protection has their own characteristics and laws, not every kind of FMCGDM model is applicable to ecological risk decision-making. Even those suitable models also need adjust some parameters to adapt to the specific decision-making problem. For example, in the evaluation and selection of those alternatives designed for paroxysmal environmental risk contingency plan, since people always can't put up with much more uncertainties in the alternatives, risk preference factor $\beta_k > 0.5$ should be more acceptable; objective weighting methods should be inclined to confirm the criteria preference; alternative preference of professional and technical staff and local people who might suffer the ecological risks should be given more respect. At the same time, if necessary, more specific parameters should be added to reflect the specific characteristics of watershed ecological risk decision-making problem.

Watershed ecological risk decision-making is a kind of decision-making problem with strong uncertainty. Except fuzzy MCDM method, the random probability theory is another core technology to handle the problems under uncertainty. Stochastic risk decision-making model is applicable to deal with the watershed ecological risk decision-making problem with external uncertainties; while fuzzy-based risk decision-making model is useful to solve ecological risk decision-making problem with internal uncertainties. This means that the fuzzy MCGDM model is more

applicable to deal with chronic ecological risk problem, while random probability theory is more applicable to address the paroxysmal ecological risk issues.

When comparing each criterion for each ecological risk management options, rough set theory can be more convenient to express the importance and dependence relationship between attributes of each criterion than fuzzy set. Therefore, it is necessary to take the combination of fuzzy set theory and rough set theory into consideration.

This article has only considered one kind of aggregation operator in determining the fuzzy weights, while it is important to notice that the operator of combining the weights with decision matrix is another key issue need to take to heart.

Even though it seems that the calculation process of the proposed fuzzy multi-criteria decision-making model is complicated, such models usually have a clear calculation procedure and relatively independent computing modules, users can depend on mathematical statistical software such as SPSS, EXCEL, and MATLAB *et al* to finish the operations. It is another important research emphasis to develop an expert system combining the computer technology and fuzzy multi-criteria decision-making theory together.

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Appendix

Judgment Submatrix

$$A_m = \begin{pmatrix} 1 & 3 & 5 & 7 & 9 \\ 1/3 & 1 & 5/3 & 7/3 & 9/3 \\ 1/5 & 3/5 & 1 & 7/5 & 9/5 \\ 1/7 & 3/7 & 5/7 & 1 & 9/7 \\ 1/9 & 3/9 & 5/9 & 7/9 & 1 \end{pmatrix}; \quad A_1 = \begin{pmatrix} 1 & 2 & 4 & 6 & 8 \\ 1/2 & 1 & 2 & 3 & 4 \\ 1/4 & 1/2 & 1 & 3/2 & 2 \\ 1/6 & 1/3 & 2/3 & 1 & 4/3 \\ 1/8 & 1/4 & 1/2 & 3/4 & 1 \end{pmatrix};$$

$$A_2 = \begin{pmatrix} 1 & 2 & 4 & 6 & 9 \\ 1/2 & 1 & 2 & 3 & 9/2 \\ 1/4 & 1/2 & 1 & 3/2 & 9/4 \\ 1/6 & 1/3 & 2/3 & 1 & 9/6 \\ 1/9 & 2/9 & 4/9 & 6/9 & 1 \end{pmatrix}; \quad A_3 = \begin{pmatrix} 1 & 2 & 4 & 8 & 8 \\ 1/2 & 1 & 2 & 4 & 4 \\ 1/4 & 1/2 & 1 & 2 & 2 \\ 1/8 & 1/4 & 1/2 & 1 & 1 \\ 1/8 & 1/4 & 1/2 & 1 & 1 \end{pmatrix};$$

$$A_4 = \begin{pmatrix} 1 & 2 & 4 & 8 & 9 \\ 1/2 & 1 & 2 & 4 & 9/2 \\ 1/4 & 1/2 & 1 & 2 & 9/4 \\ 1/8 & 1/4 & 1/2 & 1 & 9/8 \\ 1/9 & 2/9 & 4/9 & 8/9 & 1 \end{pmatrix}; \quad A_5 = \begin{pmatrix} 1 & 2 & 6 & 6 & 8 \\ 1/2 & 1 & 3 & 3 & 4 \\ 1/6 & 1/3 & 1 & 1 & 4/3 \\ 1/6 & 1/3 & 1 & 1 & 4/3 \\ 1/8 & 1/4 & 3/4 & 3/4 & 1 \end{pmatrix};$$

$$A_6 = \begin{pmatrix} 1 & 2 & 6 & 6 & 9 \\ 1/2 & 1 & 3 & 3 & 9/2 \\ 1/6 & 1/3 & 1 & 1 & 3/2 \\ 1/6 & 1/3 & 1 & 1 & 3/2 \\ 1/9 & 2/9 & 2/3 & 2/3 & 1 \end{pmatrix}; \quad A_7 = \begin{pmatrix} 1 & 2 & 6 & 8 & 8 \\ 1/2 & 1 & 3 & 4 & 4 \\ 1/6 & 1/3 & 1 & 4/3 & 4/3 \\ 1/8 & 1/4 & 3/4 & 1 & 1 \\ 1/8 & 1/4 & 3/4 & 1 & 1 \end{pmatrix};$$

$$A_8 = \begin{pmatrix} 1 & 2 & 6 & 8 & 9 \\ 1/2 & 1 & 3 & 4 & 9/2 \\ 1/6 & 1/3 & 1 & 4/3 & 3/2 \\ 1/8 & 1/3 & 3/4 & 1 & 9/8 \\ 1/9 & 2/9 & 2/3 & 8/9 & 1 \end{pmatrix}; \quad A_9 = \begin{pmatrix} 1 & 4 & 4 & 6 & 8 \\ 1/4 & 1 & 1 & 3/2 & 2 \\ 1/4 & 1 & 1 & 3/2 & 2 \\ 1/6 & 2/3 & 2/3 & 1 & 4/3 \\ 1/8 & 1/2 & 1/2 & 3/4 & 1 \end{pmatrix};$$

$$A_{10} = \begin{pmatrix} 1 & 4 & 4 & 6 & 9 \\ 1/4 & 1 & 1 & 3/2 & 9/4 \\ 1/4 & 1 & 1 & 3/2 & 9/4 \\ 1/6 & 2/3 & 2/3 & 1 & 9/6 \\ 1/9 & 4/9 & 4/9 & 6/9 & 1 \end{pmatrix}; \quad A_{11} = \begin{pmatrix} 1 & 4 & 4 & 8 & 8 \\ 1/4 & 1 & 1 & 2 & 2 \\ 1/4 & 1 & 1 & 2 & 2 \\ 1/8 & 1/2 & 1/2 & 1 & 1 \\ 1/8 & 1/2 & 1/2 & 1 & 1 \end{pmatrix};$$

$$A_{12} = \begin{pmatrix} 1 & 4 & 4 & 8 & 9 \\ 1/4 & 1 & 1 & 2 & 9/4 \\ 1/4 & 1 & 1 & 2 & 9/4 \\ 1/8 & 1/2 & 1/2 & 1 & 9/8 \\ 1/9 & 4/9 & 4/9 & 8/9 & 1 \end{pmatrix}; \quad A_{13} = \begin{pmatrix} 1 & 4 & 6 & 6 & 8 \\ 1/4 & 1 & 6/4 & 6/4 & 2 \\ 1/6 & 4/6 & 1 & 1 & 4/3 \\ 1/6 & 4/6 & 1 & 1 & 4/3 \\ 1/8 & 1/2 & 3/4 & 3/4 & 1 \end{pmatrix};$$

$$A_{14} = \begin{pmatrix} 1 & 4 & 6 & 6 & 9 \\ 1/4 & 1 & 6/4 & 6/4 & 9/4 \\ 1/6 & 4/6 & 1 & 1 & 9/6 \\ 1/6 & 4/6 & 1 & 1 & 9/6 \\ 1/9 & 4/9 & 6/9 & 6/9 & 1 \end{pmatrix}; \quad A_{15} = \begin{pmatrix} 1 & 4 & 6 & 8 & 8 \\ 1/4 & 1 & 6/4 & 2 & 2 \\ 1/6 & 4/6 & 1 & 4/3 & 4/3 \\ 1/8 & 1/2 & 3/4 & 1 & 1 \\ 1/8 & 1/2 & 3/4 & 1 & 1 \end{pmatrix};$$

$$A_{16} = \begin{pmatrix} 1 & 4 & 6 & 8 & 9 \\ 1/4 & 1 & 6/4 & 2 & 9/4 \\ 1/6 & 4/6 & 1 & 4/3 & 3/2 \\ 1/8 & 1/2 & 3/4 & 1 & 9/8 \\ 1/9 & 4/9 & 2/3 & 8/9 & 1 \end{pmatrix};$$

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